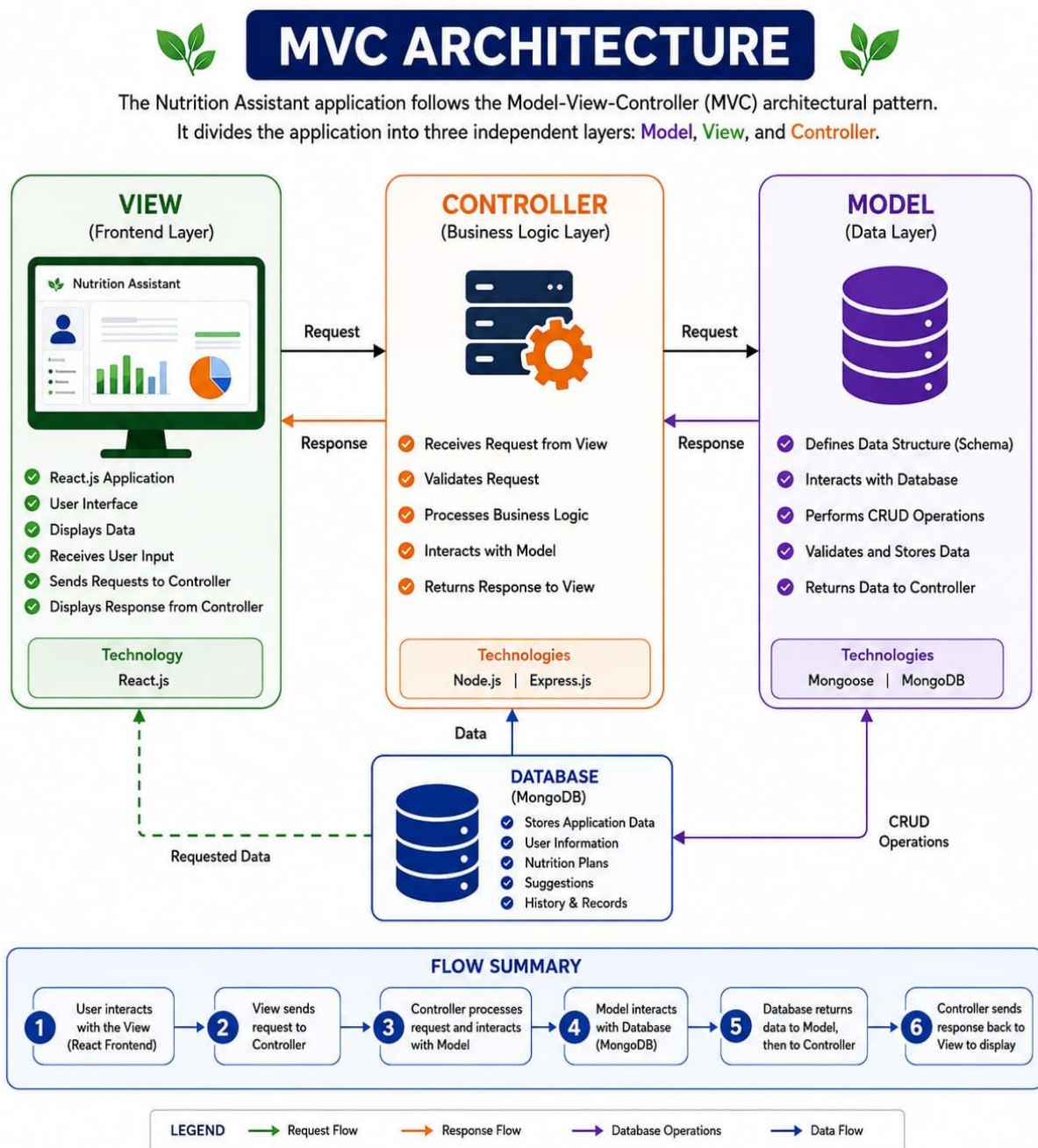


MVC PATTERN

The **Nutrition Assistant application** follows the **Model-View-Controller (MVC)** architectural pattern. MVC divides the application into three independent layers—**Model, View, and Controller**—which improves code organization, scalability, maintainability, and simplifies future enhancements.



1. Model Layer (Data Layer)

The Model Layer is responsible for managing all application data and database operations. It defines the database schema using Mongoose and communicates directly with the MongoDB database.

The Model Layer includes:

- User Model
- Suggestion Model

Responsibilities of the Model Layer

- Define MongoDB schemas
- Store user information
- Store nutrition suggestions
- Perform CRUD operations
- Maintain relationships between users and nutrition plans
- Validate data before saving

2. Controller Layer (Business Logic Layer)

The Controller Layer acts as the bridge between the View and the Model. It receives HTTP requests, processes business logic, interacts with the database, and returns appropriate responses.

- Controllers Used
- userController.js
- suggestionController.js

Responsibilities

- User Registration
- User Login
- JWT Authentication
- Profile Management
- Generate Nutrition Suggestions
- BMI Calculation
- Save Nutrition Plans
- Fetch User Suggestions
- Admin Management
- Delete Suggestions

3. View Layer (Frontend Layer)

The View Layer is developed using React.js and provides the graphical user interface. It interacts with the backend through REST APIs and displays information to users.

Pages Included

- Landing Page
- Register Page
- Login Page
- Home Dashboard
- Create Nutrition Plan
- Suggested Nutrition
- User Profile
- Admin Dashboard

Responsibilities

- Display user interface
- Collect user input
- Call backend APIs using Axios
- Display nutrition suggestions
- Handle navigation using React Router

4. Database

The application uses MongoDB as its primary database and Mongoose as the Object Data Modeling (ODM) library.

The database stores:

- User Details
- Login Credentials
- Nutrition Plans
- BMI Information
- Calorie Recommendations
- Food Suggestions
- Nutrition History
- Admin Records

The database supports efficient CRUD operations for all application modules.

5. MVC Flow in Nutrition Assistant

- The complete MVC workflow is as follows:
- User opens the React application.
- The user submits a request (Login, Register, Create Plan, etc.).
- React sends an API request to the Express server.
- Express routes the request to the appropriate Controller.
- The Controller validates the request and processes business logic.
- The Controller interacts with the MongoDB database through the Model.
- The Model retrieves or stores data in MongoDB.
- The Controller returns a JSON response.
- React receives the response and updates the user interface.
- The user views the updated nutrition information.

Advantages of Using MVC in this Project

Separation of Concerns – Each layer has a specific responsibility.

Maintainability – Easy to modify and update individual modules.

Scalability – New features can be added without affecting existing functionality.

Code Reusability – Controllers, Models, and Components can be reused.

Security – JWT authentication and middleware protect backend APIs.

Easy Testing – Each module can be tested independently.

Better Collaboration – Frontend and backend developers can work simultaneously.

Improved Performance – Organized code structure improves development efficiency and application reliability.